Design of a Class A Audio Amplifier in Common Emitter Configuration

1 Introduction

This document presents the design formulas, assertions, and calculation procedure for a Class A small-signal audio amplifier utilizing a BJT transistor in a common emitter configuration with voltage divider biasing. The objective is to establish a stable quiescent operating point (Q-point) and accurately control the voltage gain in alternating current (AC).

2 Direct Current (DC) Design Procedure

2.1 Step 1: Calculation of I_E and R_E

The design begins by setting the emitter voltage (V_E) as a percentage of the supply voltage (V_{CC}) to ensure thermal stability. A standard rule of thumb of 15% is assumed.

- Supply voltage parameter: $V_{CC} = 9$ V
- Proposed emitter voltage: $V_E = 15\%$ of $V_{CC} = 1.35$ V ≈ 1.5 V
- Fixed emitter resistance: $R_E = 1$ kΩ

The quiescent emitter current is calculated using Ohm's law:

$$I_E = \frac{V_E}{R_E} = \frac{1.5 \text{ V}}{1 \text{ k}\Omega} = 1.5 \text{ mA} \tag{1}$$

2.2 Step 2: Base Biasing (Voltage Divider)

2.2.1 Calculation of V_B

Considering the intrinsic base-emitter voltage drop of the transistor ($V_{BE} \approx 0.7$ V):

$$V_B = V_E + V_{BE} = 1.5 \text{ V} + 0.7 \text{ V} = 2.2 \text{ V} \tag{2}$$

2.2.2 Selection of the Divider Current (I_D)

To ensure that the quiescent point remains independent of the transistor's β parameter variations, the current flowing through the divider (I_D) must be at least 10 times greater than the base current (I_B).

Assuming $\beta = 200$:

$$I_B = \frac{I_C}{\beta} \approx \frac{I_E}{\beta} = \frac{1.5 \text{ mA}}{200} = 7.5\ \mu\text{A} \tag{3}$$

Establishing the divider current:

$$I_D = 10 \cdot I_B = 10 \cdot 7.5\ \mu\text{A} = 75\ \mu\text{A} \tag{4}$$

2.2.3 Calculation of the Divider Resistors (R_1 and R_2)

The base-to-ground resistor (R_2) experiences a voltage drop equal to V_B:

$$R_2 = \frac{V_B}{I_D} = \frac{2.2\text{ V}}{75\ \mu\text{A}} \approx 29.33\text{ k}\Omega \quad \rightarrow \text{(Standard value used: 30 k}\Omega\text{)} \tag{5}$$

The resistor connected to the supply (R_1) absorbs the remaining voltage ($V_{CC} - V_B$), and the current flowing through it is the divider current plus the base current ($I_D + I_B$):

$$R_1 = \frac{V_{CC} - V_B}{I_D + I_B} = \frac{9\text{ V} - 2.2\text{ V}}{75\ \mu\text{A} + 7.5\ \mu\text{A}} = \frac{6.8\text{ V}}{82.5\ \mu\text{A}} \approx 82.42\text{ k}\Omega \quad \rightarrow \text{(Standard value used: 82 k}\Omega\text{)} \tag{6}$$

3 Collector Analysis and Saturation Criterion

Initially, if a voltage gain of $A_v = 20$ is desired using the simplified formula $A_v = R_C/R_E$, the required collector resistance would be:

$$R_C = A_v \cdot R_E = 20 \cdot 1\text{ k}\Omega = 20\text{ k}\Omega \tag{7}$$

Checking the quiescent collector voltage (V_C):

$$V_C = V_{CC} - (I_C \cdot R_C) = 9\text{ V} - (1.5\text{ mA} \cdot 20\text{ k}\Omega) = 9\text{ V} - 30\text{ V} = -21\text{ V} \tag{8}$$

Design Assertion: Since the resulting voltage is negative, the transistor is completely driven into the **saturation** region. Therefore, this initial design attempt is unfeasible.

3.1 Correction of the Quiescent Point

To maximize the symmetrical output signal swing in a Class A amplifier, the collector voltage is ideally set at half of the supply voltage:

$$V_C = \frac{V_{CC}}{2} = \frac{9\text{ V}}{2} = 4.5\text{ V} \tag{9}$$

Recalculating the legitimate collector resistance to maintain the Q-point:

$$R_C = \frac{V_{CC} - V_C}{I_C} = \frac{9\text{ V} - 4.5\text{ V}}{1.5\text{ mA}} = 3\text{ k}\Omega \tag{10}$$

4 Alternating Current (AC) Analysis and Controlled Gain

If the circuit is left with a single unbypassed 1 kΩ emitter resistor, the voltage gain drops drastically to:

$$A_v \approx \frac{R_C}{R_E} = \frac{3\text{ k}\Omega}{1\text{ k}\Omega} = 3 \tag{11}$$

Conversely, if a bypass capacitor is placed in parallel with the entire emitter resistance, R_E is shorted out in AC. This causes the gain to depend solely on the transistor's intrinsic dynamic emitter resistance ($r'_e \approx 25\,\Omega$), skyrocketing the gain ($A_v > 120$) and causing severe signal clipping and distortion.

4.1 Solution: Split-Emitter Configuration

To achieve the desired gain of 20 without altering the DC quiescent point (1 kΩ total resistance), the emitter resistance is split into two series components: R_5 (unbypassed) and R_4 (bypassed in parallel with capacitor C_1).

- $R_{E\text{ Total}} = R_5 + R_4 = 150\,\Omega + 850\,\Omega = 1\text{ k}\Omega$ (Preserves the DC model).

In alternating current, capacitor C_1 acts as a short circuit across R_4, leaving only R_5 to degenerate and stabilize the AC gain.

4.2 Real World Effects and Intrinsic Dynamic Resistance

The realistic gain equation observed in the simulator incorporates the internal dynamic emitter resistance ($r'_e \approx 25\,\Omega$):

$$A_v = \frac{R_C}{R_5 + r'_e} = \frac{3000\,\Omega}{150\,\Omega + 25\,\Omega} = \frac{3000}{175} \approx 17.14 \tag{12}$$

This mathematical approximation justifies why approximately 17 units of gain are measured on the oscilloscope instead of the pure theoretical value of 20. To fine-tune the gain to exactly 20, the physical resistor R_5 can be experimentally reduced to a range between $120\,\Omega$ and $130\,\Omega$.